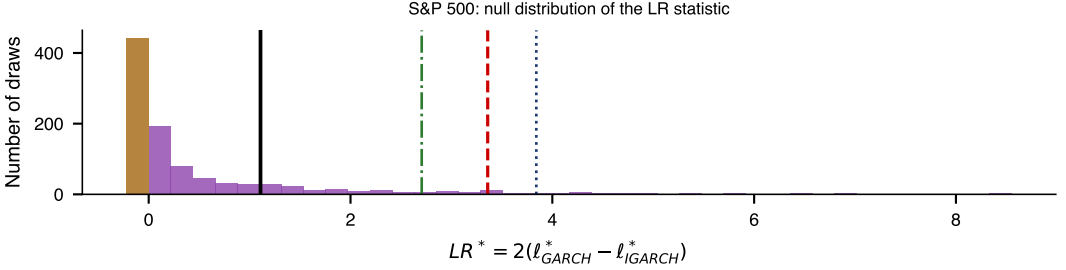
S&P 500: null distribution of the LR statistic
Number of draws
400
200
0
0
2
4
6
8
$LR^* = 2(\ell^*_{GARCH} - \ell^*_{IGARCH})$
$LR^* > 0$ on 999 paths simulated under $H_0$ (IGARCH-t)
Observed $LR$ = 1.11
Bootstrap 95% cutoff = 3.36
$\chi^2_1$ cutoff = 3.84
Mixture $\frac{1}{2}\chi^2_0 + \frac{1}{2}\chi^2_1$ cutoff = 2.71
$LR^* = 0$: 44.2% of the draws (estimate at the bound)